On the Rate Region of I.I.D. Discrete Signaling and Treating Interference as Noise for the Gaussian Broadcast Channel

Yujie Shao and Min Qiu

Global College, Shanghai Jiao Tong University

June 29, 2026

Introduction

- Future communication systems are expected to provide high spectrum efficiency and massive connectivity [1].
- Achieving these goals necessitates resource sharing among multiple users, resulting in inter-user interference.
- Successive interference cancellation (SIC) is commonly employed at the receiver to remove inter-user interference.
- SIC originates from the capacity-achieving strategies for the Gaussian broadcast channel (GBC), etc [2].

[1] E. A. Jorswieck, *Proc. IEEE*, 2024.

[2] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006.

Background

- Capacity-achieving scheme for GBC: superposition coding + Gaussian signaling + SIC [3]
- Widely adopted as the foundation of modern multiple-access designs [4].

Practical communication systems:

- Adopt discrete signaling, Gaussian signaling is difficult to realize in practice.
- SIC introduces extra decoding complexity, latency (increases with the number of users), error propagation, and privacy concerns.

[3] D. Tse and P. Viswanath, *Fundamentals of Wireless Communication*, 2005.

[4] S. M. R. Islam et al., *IEEE Commun. Surveys Tuts.*, 2017.

Literature Review

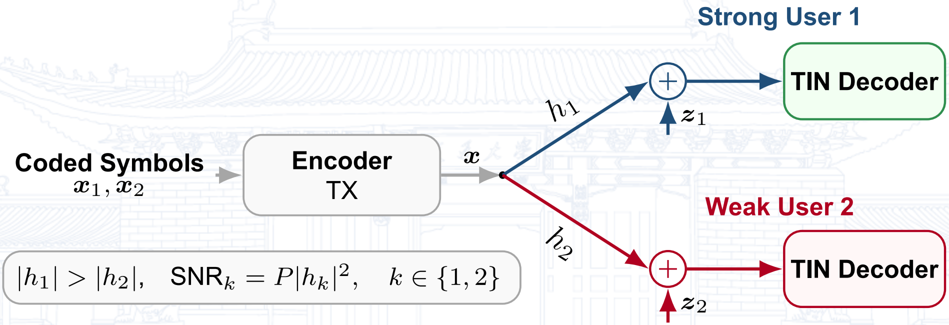
- Compared with Gaussian + SIC, discrete signaling + treating interference as noise (TIN) is more practical.
- Constant gap optimality of discrete signaling + TIN: two-user Gaussian interference channel [5], [6] and K-user GBC channels [7], [8]
- However, the constant-gap results [6]-[8] were established only at a discrete set of rate points instead of the entire achievable rate region.
- In addition, prior results, e.g., [7] and [8], rely on the linear deterministic model, which introduces approximation loss.

[5] A. Dytso et al., *IEEE Trans. Inf. Theory*, 2016.
[6] M. Qiu et al., *IEEE Trans. Inf. Theory*, 2021.
[7] S.-L. Shieh and Y.-C. Huang, *IEEE Trans. Commun.*, 2016.
[8] M. Qiu et al., *IEEE Trans. Commun.*, 2018.

Contribution

- We design and analyze new achievable schemes based on purely discrete signaling with TIN directly for the GBC without resorting to the linear deterministic model.
- We also prove that the whole achievable rate region of the proposed scheme is within a constant gap to the whole capacity region of the GBC. The gap is independent of channel parameters.
- Simulation results show that the actual gap is much smaller, and the weak user's achievable rate with PAM signaling can surpass that with Gaussian signaling in some cases.

System Model



$$\mathsf{X} = \sqrt{P}\left(\sqrt{\alpha}\mathsf{X}_1 + \sqrt{1-\alpha}\mathsf{X}_2\right),$$
$$\mathsf{Y}_k = h_k\mathsf{X} + \mathsf{Z}_k, \quad k \in \{1,2\},$$
$$\mathsf{Z}_k \sim \mathcal{N}(0,1).$$

Proposed Discrete Signaling

PAM Inputs

$$\mathsf{X}_k \overset{\text{i.i.d.}}{\sim} \text{PAM}\left(M_k, d_{\min}(\mathsf{X}_k)\right),$$

$$d_{\min}(\mathsf{X}_k) = \sqrt{\frac{12}{M_k^2 - 1}}.$$

- Modulation size constraint:

$$\prod_{i=k}^{2} M_i \le N_k = \left\lceil \sqrt{\text{SNR}_k} \right\rceil$$

Lemma 1

If $0 < \alpha \le \alpha^* \triangleq \frac{M_1^2 - 1}{M_1^2 M_2^2 - 1}$, then the minimum distance of the superimposed constellation is

$$d_{\min}(\mathsf{X}) = \sqrt{\frac{12\alpha P}{M_1^2 - 1}}.$$

- $0 < \alpha \le \alpha^*$: $d_{\min}(\mathsf{X})$ increases with α
- $\alpha^* < \alpha \le 1$: $d_{\min}(\mathsf{X})$ fluctuates and can drop to 0

Main Result: Constant Gap

Definition 1

An achievable region is within a constant gap $\Delta > 0$ to the capacity region if for any rate pair (R_1, R_2) in the capacity region, $(R_1 - \Delta, R_2 - \Delta)$ is achievable.

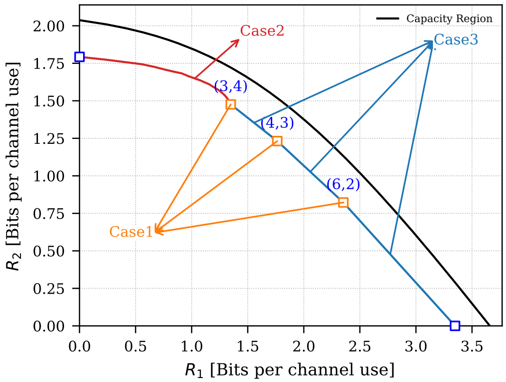
Theorem 1

The following rate region $\mathcal{R}$ is achievable with the proposed PAM signaling and TIN decoding, and is within a constant gap to the GBC capacity $\mathcal{C}_{\mathsf{GBC}}$.

$$\mathcal{R} \triangleq \bigcup_{\substack{\alpha\in[0,\alpha^*] \\ \prod_{i=k}^{2} M_i \le N_k,\ k\in\{1,2\}}} \left\{ 0 \le R_k \le I(\mathsf{X}_k;\mathsf{Y}_k) \right\} \bigcup \mathcal{R}_{\mathsf{TS}},$$

$$\mathcal{R}_{\mathsf{TS}} \triangleq \text{co} \left(\bigcup_{\substack{\alpha\in\{\alpha^*,1\} \\ \prod_{i=k}^{2} M_i \le N_k,\ k\in\{1,2\}}} \left\{ 0 \le R_k \le I(\mathsf{X}_k;\mathsf{Y}_k) \right\} \right).$$

Proof Sketch of Main Result



Define $\Delta_k(\alpha) \triangleq C_k(\alpha) - I(\mathsf{X}_k; \mathsf{Y}_k)$ for $k \in \{1, 2\}$.

The proof consists of three cases:

- Case 1: $\Delta(\alpha)$ is a constant for $\alpha = \alpha^*$
- Case 2: $\Delta(\alpha)$ is a constant for $0 < \alpha < \alpha^*$
- Case 3: Constant gap between $\mathcal{R}_{\mathsf{TS}}$ and the corresponding $\mathcal{C}_{\mathsf{GBC}}$

The single-user cases, i.e., $\alpha = 0$ and $\alpha = 1$, are trivial and omitted.

For a detailed proof, see the extended arXiv version: arXiv:2604.04092.

Mutual Information Lower Bound for $\alpha \in (0, \alpha^*]$

Achievable Rate of User 1: By using Ozarow-Wyner-B bound in [5],

$$
\begin{aligned}
I(\mathsf{X}_1;\mathsf{Y}_1) &\geq \log_2 M_1 - \frac{1}{2}\log_2\left(\tfrac{2\pi e}{12}\right) - \frac{1}{2}\log_2\left(1 + \frac{12}{d_{\mathsf{min}}(h_1\mathsf{X})^2}\right) \\
&\geq \frac{1}{2}\log_2\left(M_1^2 + \alpha\mathsf{SNR}_1 + \frac{M_1^2 - 1}{\alpha\mathsf{SNR}_1}\right) - \log_2 M_1 - \frac{1}{2}\log_2\left(\tfrac{2\pi e}{12}\right).
\end{aligned}
$$

Achievable Rate of User 2: Define $\mathsf{Z}_2' \triangleq \frac{\mathsf{Z}_2 + \sqrt{\alpha\mathsf{SNR}_2}\mathsf{X}_1}{\sqrt{1+\alpha\,\mathsf{SNR}_2}}$, and construct the corresponding equivalent channel $\mathsf{Y}_2' = \mathsf{X}_2' + \mathsf{Z}_2'$. Similarily, by using Ozarow-Wyner-B bound in [5],

$$
I(\mathsf{X}_2;\mathsf{Y}_2) = I(\mathsf{X}_2';\mathsf{Y}_2') \geq \log_2 M_2 - \frac{1}{2}\log_2\left(\tfrac{2\pi e}{12}\right) - \frac{1}{2}\log_2\left(1 + \frac{1+\alpha\,\mathsf{SNR}_2}{(1-\alpha)\mathsf{SNR}_2}(M_2^2 - 1)\right).
$$

Note: d_{min} is a function of α,SNR and M.

[5] A. Dytso et al., *IEEE Trans. Inf. Theory*, 2016.

Case 1: $\alpha = \alpha^*$

Set $M_1 M_2 = N_1 = \left\lceil \sqrt{\mathsf{SNR}_1} \right\rceil$ to use the largest possible modulation size. Moreover, set $M_2 = \left\lfloor \frac{N_1}{M_1} \right\rfloor$ to fulfill the above constraint.

- **User 1:** Let $\mathsf{SNR}_1 = (M_1^2 M_2^2 - 1)\delta$ for some constant $\delta > 0$. Then,

$$\Delta_1(\alpha^*) \le \frac{1}{2}\log_2\left(1 + \frac{M_1^2 - 1}{M_1^2}\delta + \frac{1}{M_1^2\delta}\right) + \frac{1}{2}\log_2\left(\frac{2\pi e}{12}\right) < 1.188.$$

- **User 2:**

$$\Delta_2(\alpha^*) \le \frac{1}{2}\log_2\left(1 + \frac{1}{M_2^2}\frac{(M_1^2 M_2^2 - M_1^2)\,\mathsf{SNR}_2}{(M_1^2 M_2^2 - 1) + (M_1^2 - 1)\mathsf{SNR}_2}\right.$$
$$\left. + \frac{(M_1^2 M_2^2 - 1) + (M_1^2 - 1)\mathsf{SNR}_2}{M_1^2 M_2^2 \mathsf{SNR}_2}\right) + \frac{1}{2}\log_2\left(\frac{2\pi e}{12}\right) < 1.175.$$

Case 2: $\alpha \in (0, \alpha^*)$

Set $M_2 = N_2$ and $M_1 = \left\lfloor \frac{N_1}{M_2} \right\rfloor$ to approach the capacity region.

- **User 1:** $\Delta_1(\alpha) < C_1(\alpha) < 1$ when $\alpha\mathsf{SNR}_1 \leq 1$. So consider $\alpha\mathsf{SNR}_1 > 1$,

$$\Delta_1(\alpha) < \tfrac{1}{2}\log_2\left(M_1^2 + \alpha\mathsf{SNR}_1 + \frac{M_1^2 - 1}{\alpha\mathsf{SNR}_1}\right) - \log_2(M_1) + \tfrac{1}{2}\log_2\left(\tfrac{2\pi e}{12}\right) < 1.188.$$

- **User 2:** $\Delta_2(\alpha) < C_2(\alpha) < 1.161$ for $\mathsf{SNR}_2 \leq 4$. So consider $\mathsf{SNR}_2 > 4$,

$$\Delta_2(\alpha) \leq \tfrac{1}{2}\log_2\left(1 + \frac{(1-\alpha)\mathsf{SNR}_2}{(1+\alpha\,\mathsf{SNR}_2)M_2^2} + \frac{1+\alpha\,\mathsf{SNR}_2}{(1-\alpha)\mathsf{SNR}_2}\,\frac{M_2^2 - 1}{M_2^2}\right) + \tfrac{1}{2}\log_2\left(\tfrac{2\pi e}{12}\right) < 0.839.$$

Case 3: Time-sharing Rate Region

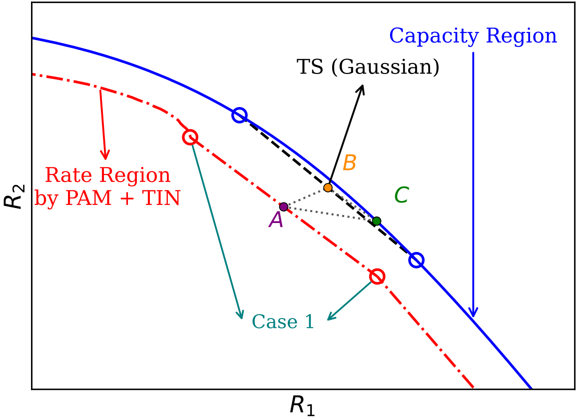


Illustration of achievable rate points.

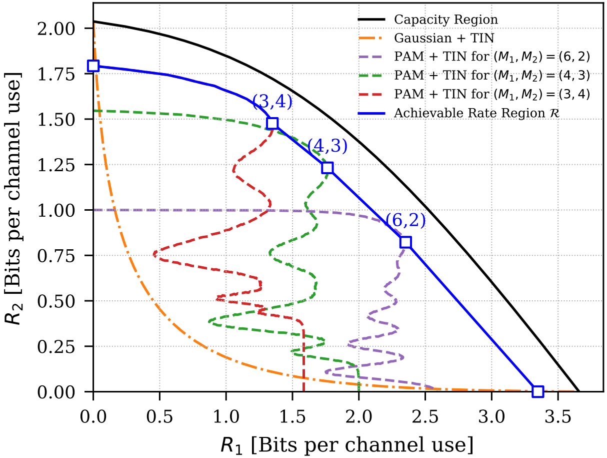
Proof Steps:

- Prove $\Delta_{1,\mathrm{AB}} < 1.188$ and $\Delta_{2,\mathrm{AB}} < 1.175$ following Case 1.
- Prove $\Delta_{1,\mathrm{BC}} < 1.772$ and $\Delta_{2,\mathrm{BC}} < 2.985$.
- Using the triangular inequality, we have

$$\Delta_{1,\mathrm{AC}} < |\Delta_{1,\mathrm{AB}}| + |\Delta_{1,\mathrm{BC}}| < 2.960,$$

$$\Delta_{2,\mathrm{AC}} < |\Delta_{2,\mathrm{AB}}| + |\Delta_{2,\mathrm{BC}}| < 4.160.$$

Numerical Results: Achievable Rate Region



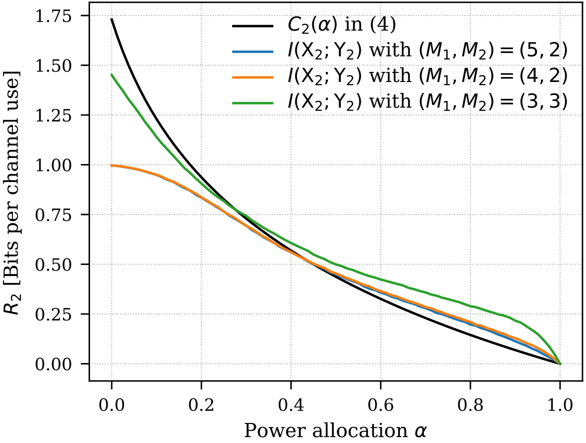
Setup

$$(\mathrm{SNR}_1, \mathrm{SNR}_2) = (22, 12)\ \mathrm{dB}$$

$$(M_1, M_2) = (6, 2), (4, 3), (3, 4)$$

- The actual gap is much smaller than the upper bound of the analytical gap.

Numerical Results: Weak User's Rate



Setup

$$(\mathrm{SNR}_1, \mathrm{SNR}_2) = (20, 10)\ \mathrm{dB}$$

$$(M_1, M_2) = (5, 2), (4, 2), (3, 3)$$

- The weak user's rate R_2 is larger than C_2 for some cases.

Conclusion

- We directly designed and analyzed our scheme for the GBC without using the linear deterministic model, thereby leading to simpler designs and proofs.
- We have proven that the whole rate region achieved by the proposed scheme is within a constant gap to the capacity region.
- In addition, we showed that the weak user's rate achieved by PAM can be larger than that by Gaussian signaling in some cases.
- As part of future work, we plan to extend the proposed design and analysis to the GBC with more than two users.

Thank You